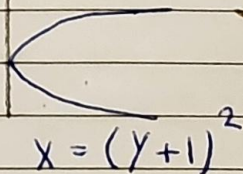
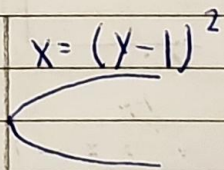
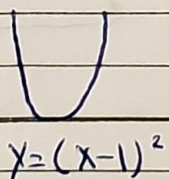
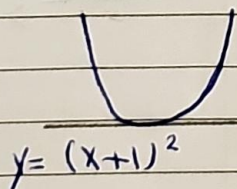
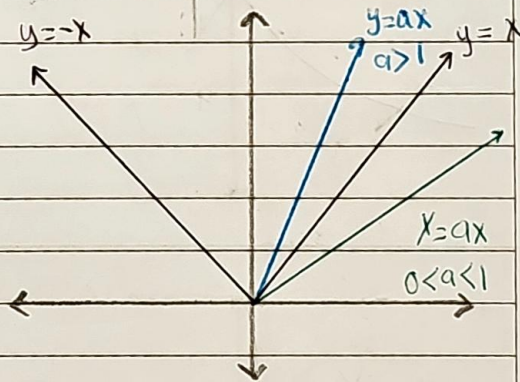
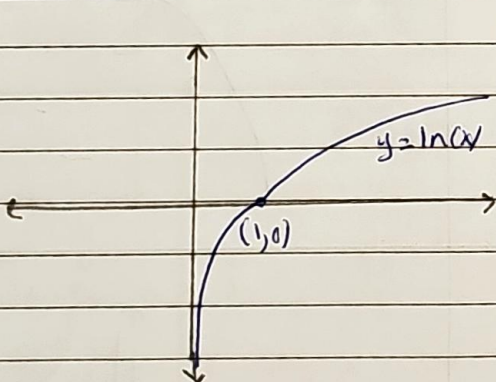
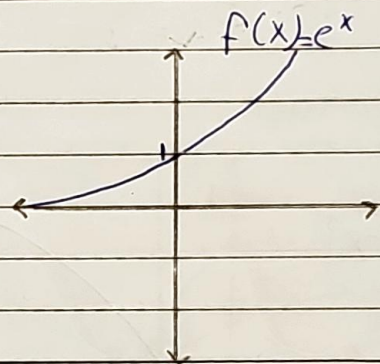
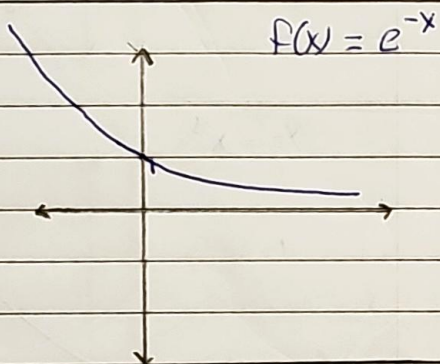
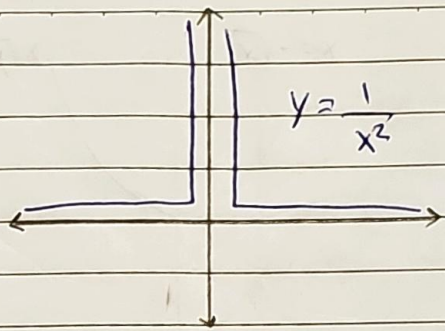
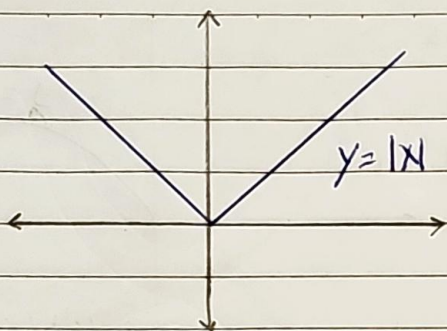


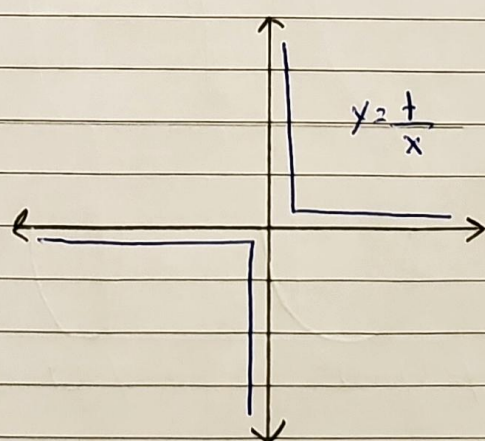
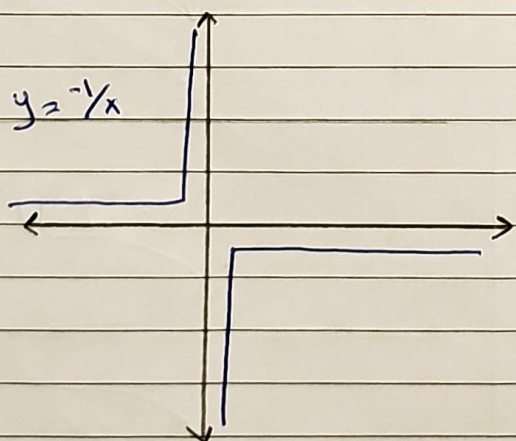
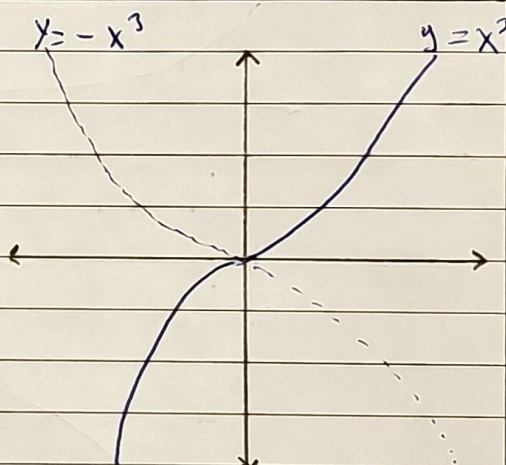
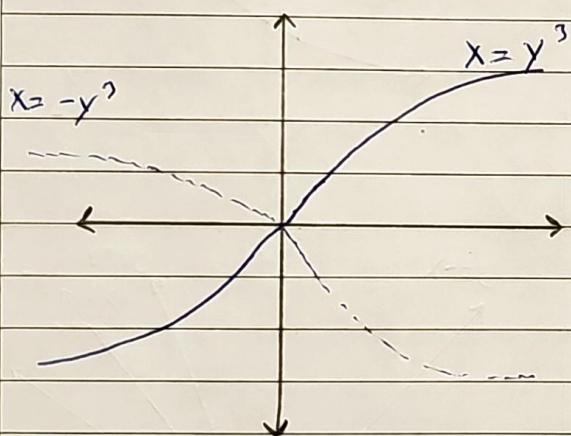
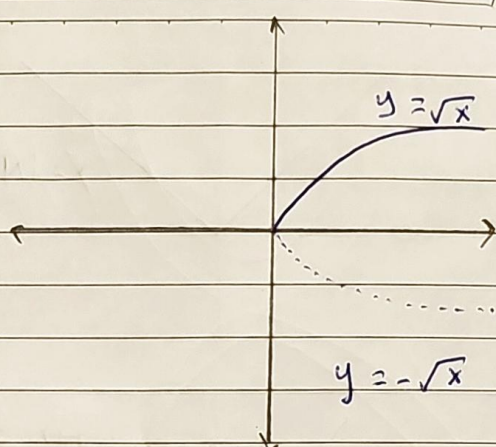
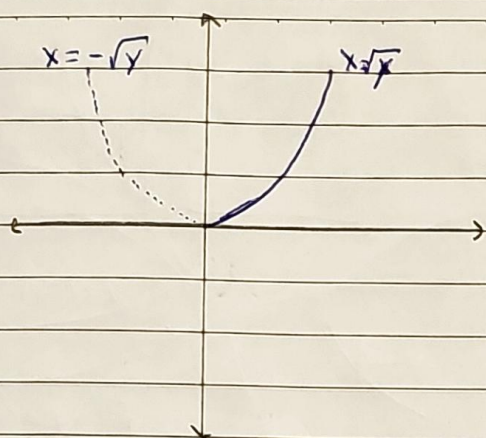
5

الرسائل

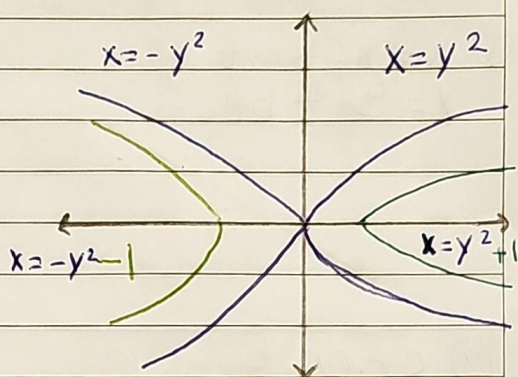
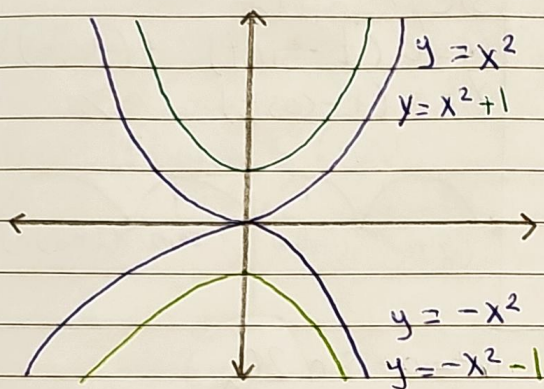


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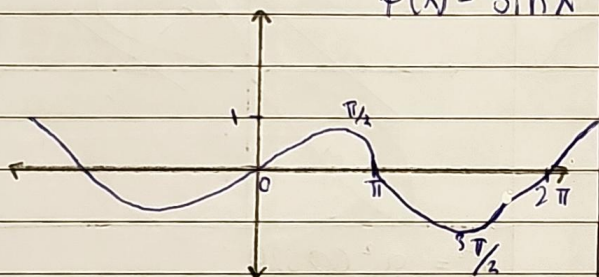
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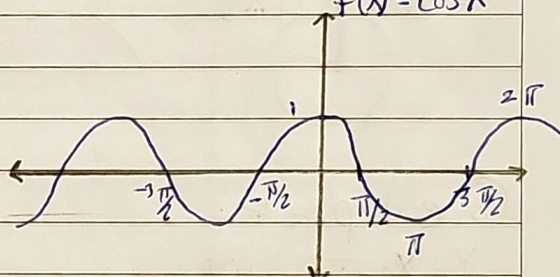
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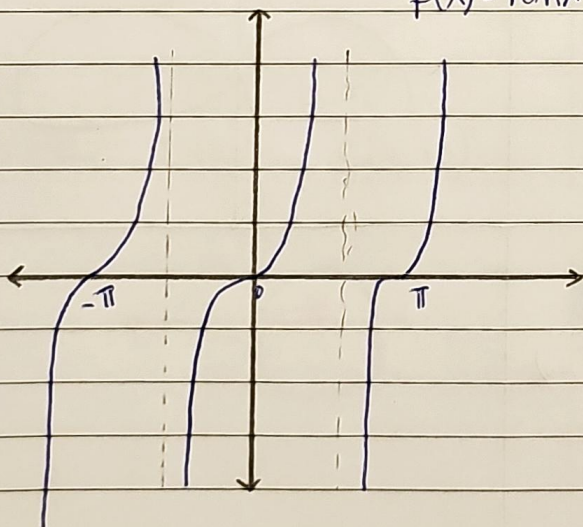
$f(x) = \sin x$



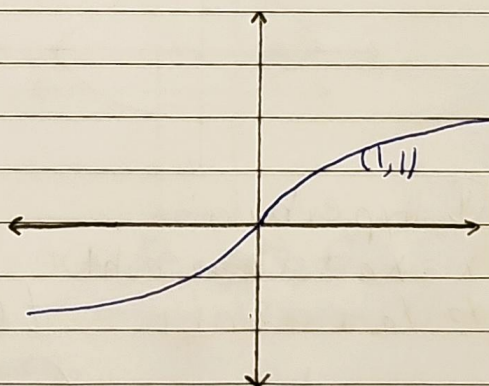
$f(x) = \cos x$



$f(x) = \tan x$

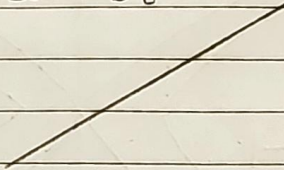


$f(x) = \tan^{-1} x$



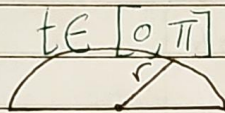
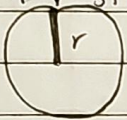
10.1 + 10.2

① $x = x_0 + at$
 $y = y_0 + bt$



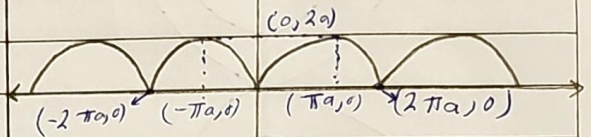
② Circle:

$x = x_0 + r \cos t$
 $y = y_0 + r \sin t$

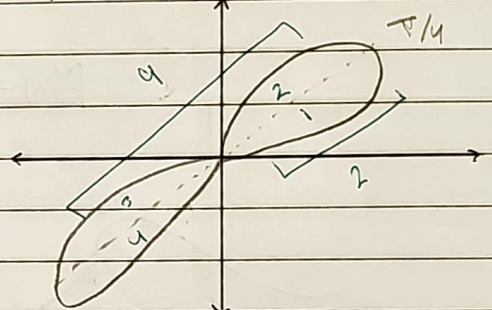


⑤ Cycloid:

$x = a(t - \sin t)$ $t \in (-\infty, \infty)$
 $y = a(1 - \cos t)$ $a > 0$

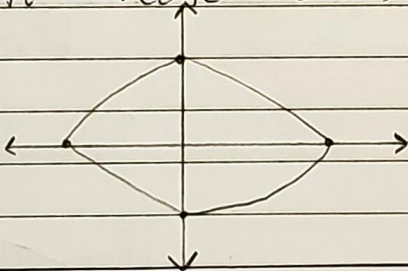


$r^2 = a^2 \sin 2\theta$

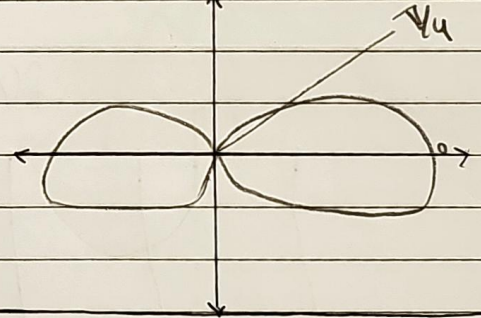


③ ellipse:

$x = x_0 + a \sin t$
 $y = y_0 + a \cos t$ $t \in (-\infty, \infty)$

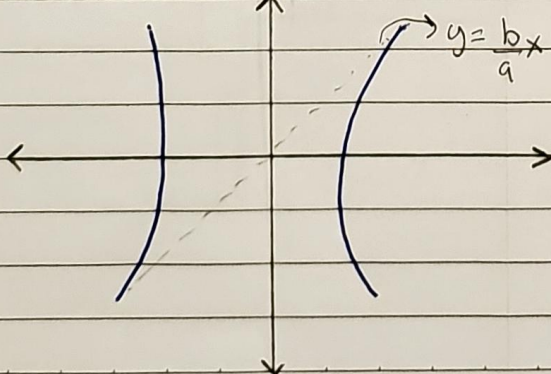


$r^2 = a^2 \cos 2\theta$



④ Hyperbola:

$x = x_0 + a \cosh t$
 $y = y_0 + a \sinh t$ $t \in (-\infty, \infty)$

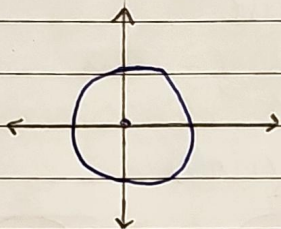


5

10.3 + 10.4

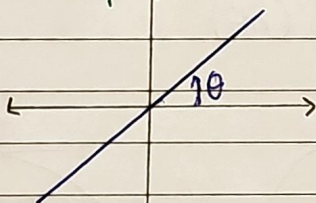
$$① r = c ; c \neq 0$$

$$\text{radius} = |c| \quad C(0,0)$$



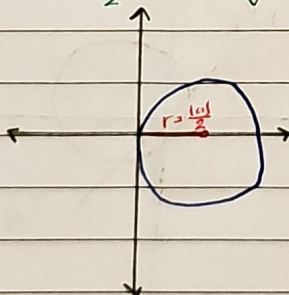
$$② \theta = c ; c \neq 0$$

line passes through the pole



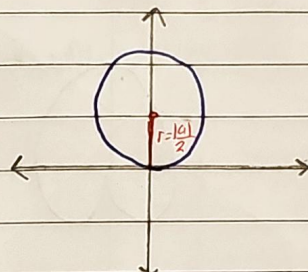
$$③ r = a \cos \theta ; a \neq 0$$

$$\text{circle: } r = \frac{|a|}{2}, C\left(\frac{a}{2}, 0\right)$$



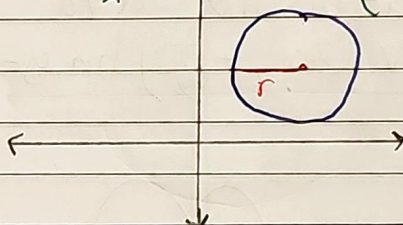
$$④ r = a \sin \theta \quad a \neq 0$$

$$\text{radius} = \frac{|a|}{2}, C\left(0, \frac{a}{2}\right)$$



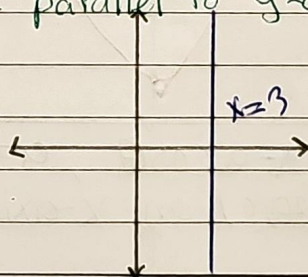
$$⑤ r = a \cos \theta + b \sin \theta \quad a, b \neq 0$$

$$\text{radius} = \frac{1}{2} \sqrt{a^2 + b^2} \quad C\left(\frac{a}{2}, \frac{b}{2}\right)$$



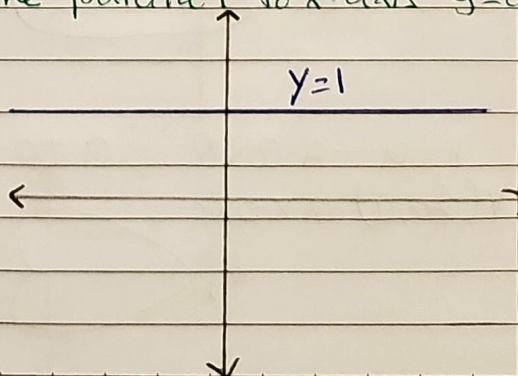
$$⑥ r = a \sec \theta, a \neq 0, \theta \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$$

line parallel to y-axis $x = a$



$$⑦ r = a \csc \theta, a \neq 0, \theta \in (0, \pi)$$

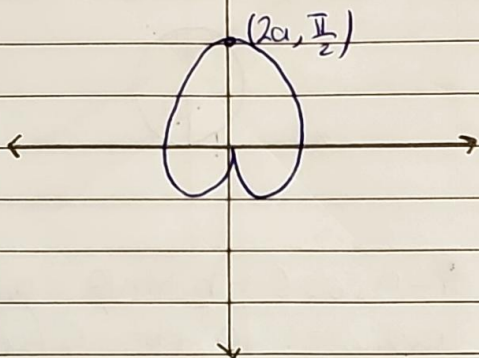
line parallel to x-axis $y = a$



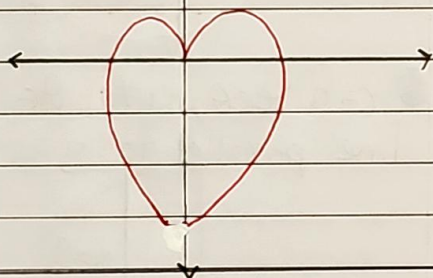
Cardioid

rose

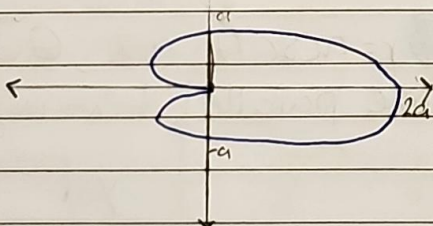
8 $r = a + a \sin \theta$ $a \neq 0$
 $V(2a, \frac{\pi}{2})$ on y-axis



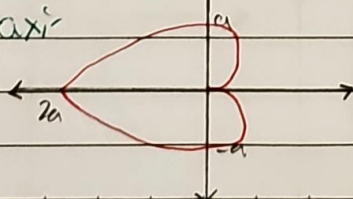
$r = a - a \sin \theta$ $a \neq 0$
 $V(2a, -\frac{\pi}{2})$ on y-axis



9 $r = a + a \cos \theta$ $a \neq 0$
 $V(2a, 0)$ on x-axis



$r = a - a \cos \theta$ $a \neq 0$
 $V(-2a, 0)$ on x-axis



10 $r = a \cos(n\theta)$ $a \neq 0$
 $= a \sin(n\theta)$ $n \geq 2$

rose

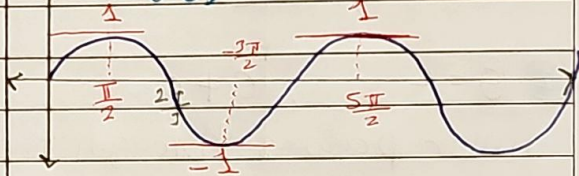
n odd
 n leaves

n even
 2n leaves

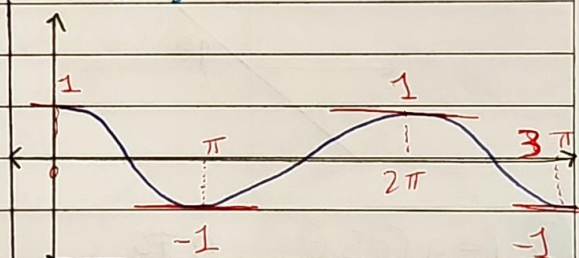
① $r = \pm a$

② $\cos(n\theta), \sin(n\theta) = \pm 1$

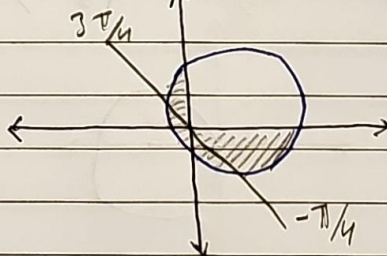
$\sin(n\theta)$



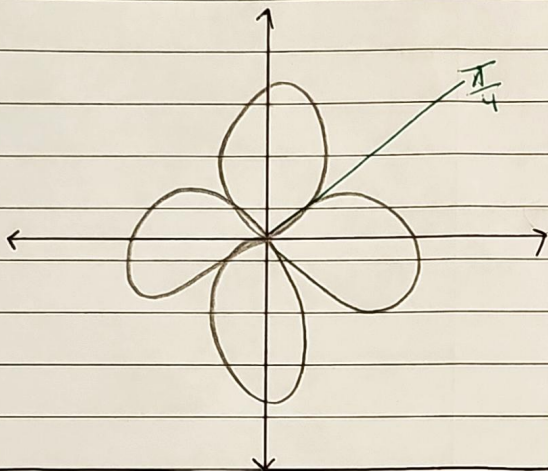
$\cos(n\theta)$



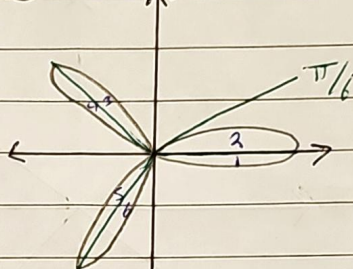
$r = \sin \theta + \cos \theta$



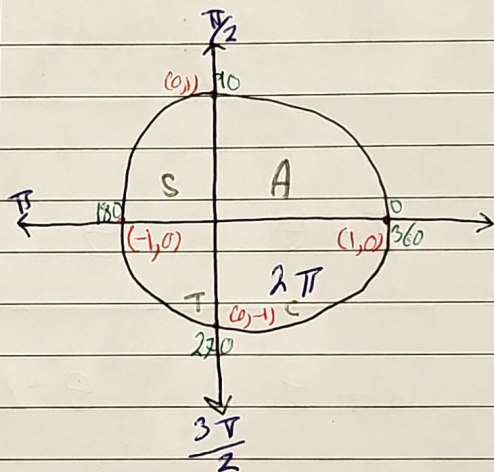
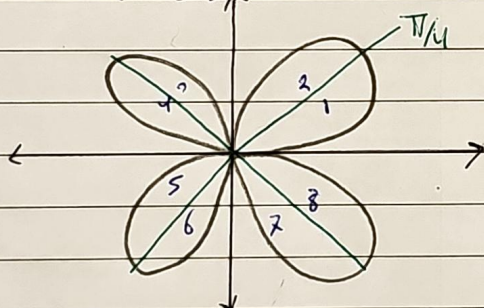
$$r = a \cos(2\theta)$$



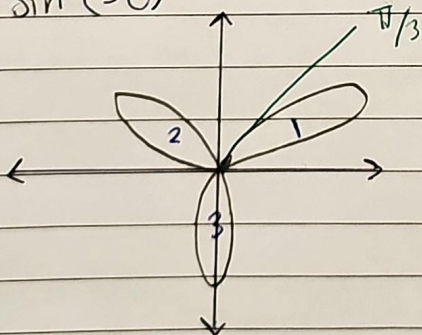
$$r = a \cos(3\theta)$$

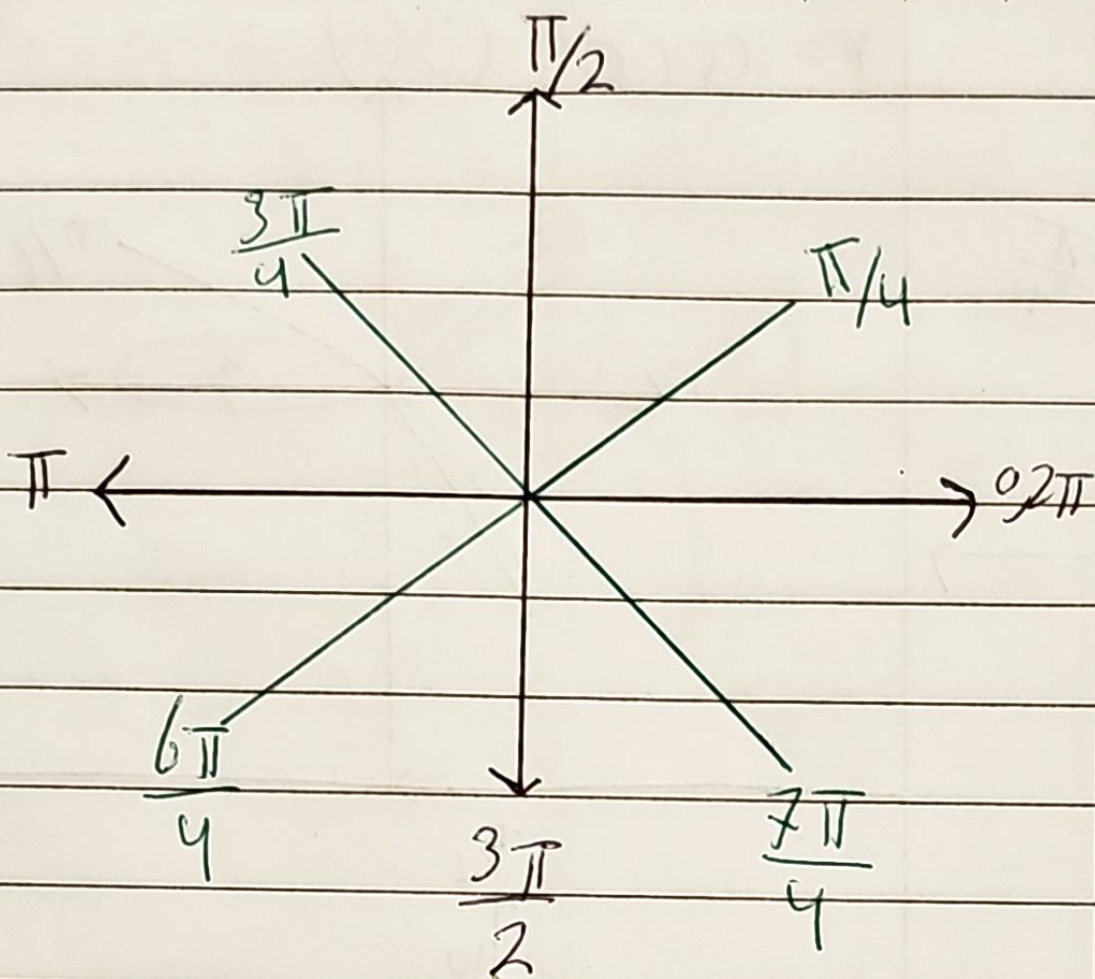


$$r = a \sin(2\theta)$$



$$r = a \sin(3\theta)$$





$$\pi - \theta$$

$$\theta$$

$$\pi + \theta$$

$$\theta - 2\pi$$